

Fractal-Domain Vision Graph Neural Network for Remote Sensing Ground Target Classification

Jiacheng Yin , Tao Zhen , Gang Xiong , and Wenxian Yu , *Senior Member, IEEE*

Abstract—To the best of our knowledge, this paper is the first to integrate fractal signal processing with vision graph neural networks, establishing a new graph representation learning paradigm consistent with fractal dynamics. Building on this foundation, we propose a *Fractal-domain Vision Graph Neural Network (FD-ViG)*. Specifically, FD-ViG includes: (i) a *Fractal-Domain Learning Module* that maps images into the fractal-domain using local Hölder exponents and the Singularity Power Spectrum (SPS), enabling fractal-spatial feature fusion; (ii) a *Fractal Graph Construction Module* that adaptively generates a topology by combining semantic attention with fractal similarity in the fractal feature space; and (iii) a *Graph Propagation Module* with power-law multi-scale propagation to realize cross-scale diffusion and aggregation, enabling coupled texture-structure learning. Experiments on UCMerced, RSSCN7, and SIRI-WHU achieve overall accuracies of 91.75%, 89.52%, and 92.78%, respectively. Compared with representative vision graph models such as ViG, WiGNet, and ViHGNN, our method achieves consistent improvements over prior methods across all three datasets, while remaining lightweight (2.6 M parameters). Moreover, despite having far fewer parameters than ResNet-18, our model yields competitive or better performance on two datasets, and further demonstrates strong generalization ability in cross-dataset evaluation on SAR imagery. This work provides a principled and effective bridge between fractal theory and graph deep learning, benefiting interpretable remote sensing scene understanding under complex textures and structures.

Index Terms—Stochastic fractals, fractal graph neural networks, fractal attention, fractal graph topology, remote sensing.

I. INTRODUCTION

WITH the continuous improvement of remote sensing spatial resolution and acquisition capability, remote sensing scenes exhibit increasingly complex geometric shapes, scale variations, and texture/structural patterns. These factors make interpretation harder due to cross-scale texture distributions, irregular spatial relations, and pronounced intra-class heterogeneity, which jointly degrade feature stability and discriminability. Under this setting, models must capture both local texture complexity and global geometric topology to overcome the modeling bottleneck caused by scale changes and irregular spatial organization. How to jointly model texture cues and

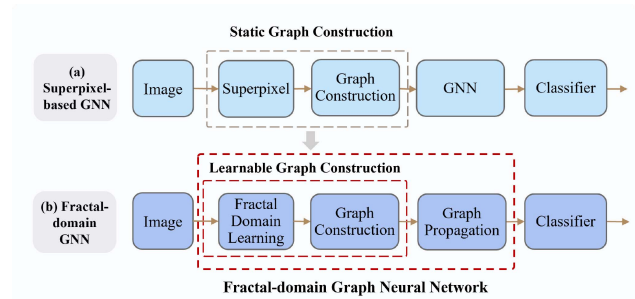


Fig. 1. Comparison between a classic “superpixel segmentation + graph propagation” pipeline (top) and our fractal-domain graph modeling pipeline (bottom). Our core modules include fractal-domain learning, fractal graph construction, and power-law graph propagation.

spatial/structural relations *while preserving interpretability* becomes a central problem in remote sensing scene classification.

Convolutional neural networks (CNNs) rely on local kernels and are often limited in capturing global geometric relations, especially under significant scale variation and complex structures. Vision Transformers (ViTs) [1] model long-range dependencies via self-attention, yet they usually lack explicit spatial organization constraints. For high-resolution remote sensing imagery, this may lead to insufficient structural priors and unstable topology-aware associations.

To enhance spatial relation modeling, graph neural networks (GNNs) [2], [3] have been introduced into remote sensing interpretation, where feature propagation over graph topology captures region-level dependencies. However, graph construction commonly depends on superpixel/region segmentation strategies such as Simple Linear Iterative Clustering [4]. Since such segmentation is driven by color consistency and local edge shape, it can generate distorted partitions and unstable adjacency under complex textures, multi-scale changes, or illumination differences. Errors then propagate to subsequent message passing and degrade performance.

To reduce reliance on external segmentation priors, Vision GNN (ViG) [5] constructs graphs adaptively in feature space via attention, allowing topology to evolve with semantics. Nevertheless, its graph construction is largely based on semantic similarity. In remote sensing, many regions exhibit strong spatial organization patterns (e.g., regular building layouts, directional continuity of runways/roads, clustered distribution of ships in harbors). Semantic similarity alone can be insufficient to reflect these structural patterns [6], leaving room for improvement in multi-scale structure modeling.

Received 6 January 2026; revised 28 March 2026; accepted 24 April 2026. Date of publication 5 May 2026; date of current version 6 August 2026. This work was supported in part by the Shanghai Natural Science Foundation of China under Grant 257R1401197) and in part by the National Natural Science Foundation of China under Grant 62071293. Recommended for acceptance by B. Rosenhahn. (Corresponding author: Gang Xiong.)

The authors are with the School of Automation and Intelligent Sensing, Shanghai Jiao Tong University, Shanghai 200240, China (e-mail: gxiong@sjtu.edu.cn). Digital Object Identifier 10.1109/TPAMI.2026.3690544

Meanwhile, fractal signal processing indicates that many remote sensing textures exhibit statistical self-similarity and power-law energy distributions. Local Hölder exponents and Singularity Power Spectrum Density (SPSD) [7], [8], [9] can stably characterize texture roughness and structural complexity, showing robustness to noise and scale changes in optical and SAR tasks. Recent FracFormer [10] embeds singularity-domain features into Transformers and improves representation and classification under complex textures. However, fractal-domain methods mainly enhance feature representation and do not explicitly model topological dependencies among regions. In other words, the fractal-domain can tell *which regions are similar in texture*, but not *how these regions should be organized topologically*. As illustrated in Fig. 1, this motivates a graph representation paradigm that injects fractal-domain structural cues into topology construction.

Therefore, a key challenge is to incorporate both semantic similarity and fractal multi-scale structural information during graph construction, so that the resulting graph representation captures multi-scale texture consistency and spatial structural relations.

Main Contributions:

- *Unified fractal-domain graph learning architecture.* We build an end-to-end framework from fractal feature extraction to graph propagation.
- *Fractal-based graph construction and propagation.* We propose an adaptive graph generation mechanism that fuses semantic attention and fractal similarity in the fractal feature space, and a power-law multi-scale propagation scheme for consistent topology and cross-scale diffusion.
- *A lightweight fractal-domain vision GNN for remote sensing classification.* We design a compact model (2.6 M parameters) and achieve strong performance on UCMerced, RSSCN7, SIRI-WHU, and OpenSARUrban validating the stability and interpretability of fractal-domain propagation for complex scene recognition.

The remainder of this paper is organized as follows. Section III details the overall architecture and key modules, including fractal-domain feature extraction, graph construction, and power-law multi-scale propagation. Section IV analyzes convergence properties of SPSP estimation and stability of power-law propagation. Section V presents experiments and visualization analyses. Section VI concludes the paper.

II. RELATED WORK

A. Visual GNNs and Learnable Graph Construction

Graph neural networks (GNNs) introduce explicit structural modeling into vision tasks via message passing over nodes and edges, enabling relation-aware representation learning beyond local convolutional receptive fields [2], [11], [12]. Early visual graph pipelines often relied on hand-crafted graph construction, e.g. building graphs from superpixels or region partitions [4], [13], which makes the topology sensitive to low-level appearance changes and segmentation errors. To reduce dependence on fixed adjacency or external priors, recent work has shifted

toward *learnable* graph construction. Vision GNN (ViG) represents an image as a graph of patch tokens and dynamically constructs adjacency in the feature space via similarity-based retrieval, allowing the topology to evolve with semantic representations [5]. Building on this direction, Elsharkawi's thesis further revisits the construction mechanism and introduces learnable reparameterization to improve controllability and robustness of adjacency generation, mitigating instability under noise and complex scenes [14]. Overall, these methods upgrade graph topology from a fixed prior to an end-to-end learnable object, strengthening the expressiveness of vision graph models.

B. From Pairwise Edges to High-Order Structures: Hypergraph-Based Vision Modeling

Despite improved flexibility, ViG-style methods still primarily model *pairwise* relations between patches. However, many visual scenes exhibit group-level or region-set structures that are difficult to fully express with only pairwise edges. Hypergraphs provide a natural extension by introducing hyperedges that connect multiple nodes simultaneously, enabling high-order interaction modeling [15]. In the vision domain, Vision HGNN incorporates hyperedge structures into patch-based graph modeling so that multiple patches can be associated through shared hyperedges, better capturing group-wise context and compositional patterns [16]. More recently, Hypergraph Vision Transformers further integrate hypergraph structures with Transformer-style representations, exploring hierarchical or bipartite hypergraph formulations to express more complex semantic and structural dependencies [17]. These advances suggest that moving from pairwise relations to high-order interactions is a promising direction for richer structural representation.

C. Graph Modeling in Remote Sensing Scene Understanding

In remote sensing, graph modeling is often used to complement CNNs and Transformers in capturing spatial organization and region-to-region dependencies. Earlier remote sensing scene classification methods mainly focused on topic-based or feature-fusion paradigms [18], [19], while graph-based approaches increasingly emphasize relational reasoning over regions or patches. Representative hybrid frameworks combine CNN feature extraction with graph propagation to exploit both local appearance and spatial relations, such as multi-output architectures that jointly learn CNN and GNN pathways for remote sensing scene classification [20]. This line of research reflects a transition from segmentation-dependent graph priors to more integrated structure learning; nevertheless, how to inject *stable and physically meaningful* texture/structure cues into graph construction and propagation remains insufficiently explored in remote sensing scenarios.

D. Fractals and Hölder Exponents in Remote Sensing and Learning Systems

Fractal theory provides an effective tool for characterizing complex textures in remote sensing imagery. Local Hölder exponents quantify local regularity and roughness variations and have

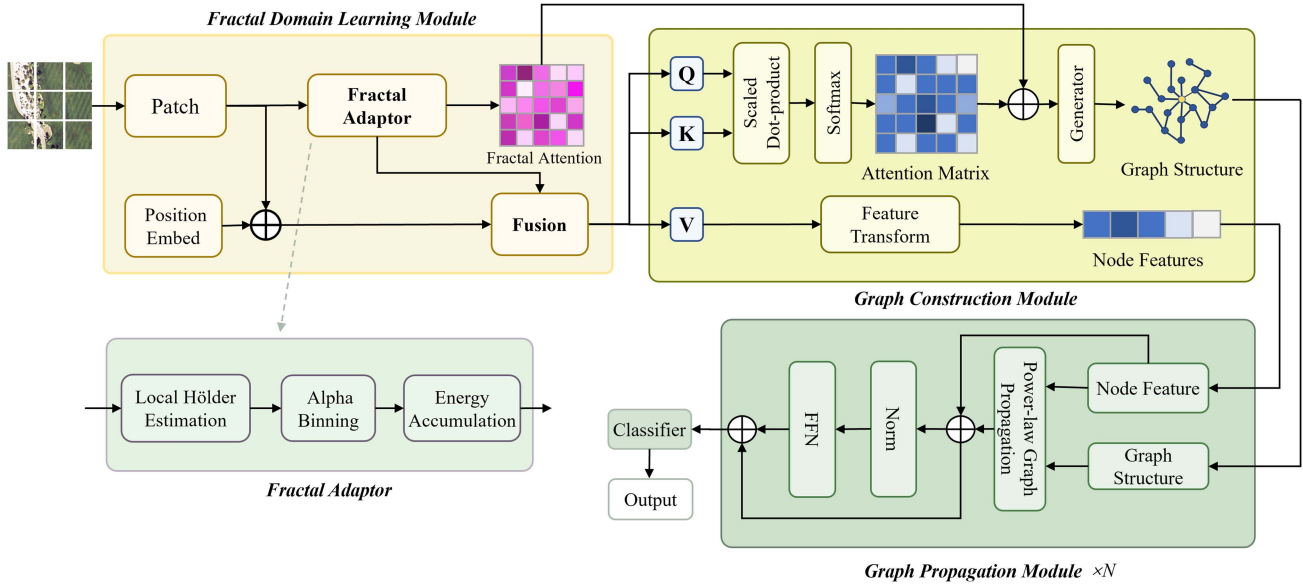


Fig. 2. Overall architecture of FD-ViG. The pipeline mainly consists of Fractal-Domain Learning, Graph Construction, and Graph Propagation.

been applied to high-resolution satellite image segmentation and structural analysis [21], as well as fine-grained tasks such as individual tree detection from UAV imagery, where they help capture subtle texture differences [22]. Meanwhile, efforts have also explored optimization and learning of Hölder-based descriptors and estimators, including genetic-algorithm-based optimization of Hölder image descriptors [23] and genetic-programming-based evolution of pointwise Hölder exponent estimators [24]. Overall, these studies indicate that Hölder/fractal features are physically interpretable and can provide relatively stable cues for describing texture roughness and structural complexity.

In terms of combining fractals with deep learning, FractalNet introduces self-similar branching structures as an architectural inductive bias to build ultra-deep networks [25]. However, it mainly leverages *fractal-like network morphology* rather than explicitly incorporating fractal modeling procedures or spectrum-based representations. To better exploit fractal-spectrum modeling, Fractal Filter Networks (FFN) construct learnable spectrum-related layers and fractal filter layers to replace self-attention blocks in Transformers, improving radar sea clutter based sea-state classification [26]. Furthermore, fractal-domain representations have been more tightly integrated with Transformer architectures: Xiong et al. propose a fractal-domain Transformer framework that embeds fractal-domain representations into attention-based modeling for SAR ship classification [10], demonstrating that fractal features can serve not only as traditional descriptors but also as intermediate representations within end-to-end learning systems. Related fractal/singularity-domain signal processing studies further support the interpretability and robustness of such representations [7], [8], [9], [27].

E. Positioning of This Work

Existing studies advance vision structural modeling along multiple directions, including learnable graph construction [5],

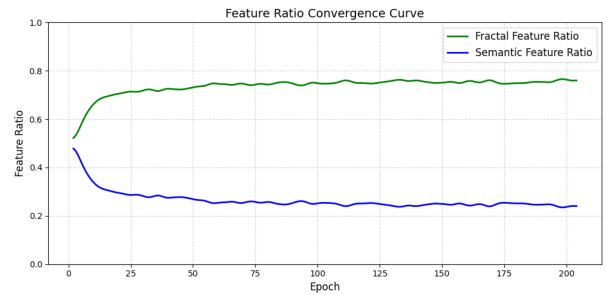


Fig. 3. Fractal feature ratio during training. The curve stabilizes after about 30 epochs, indicating that fractal features gradually dominate the fused representation.

[14], high-order hypergraph reasoning [16], [17], and fractal-domain modeling [10], [21], [22], [25], [26]. However, these directions have not been sufficiently unified for remote sensing scene understanding: learnable vision graphs often prioritize semantic similarity, while fractal-domain methods mainly strengthen texture representation without explicitly shaping graph topology and propagation. In contrast, our FD-ViG injects fractal-domain similarity as a complementary cue to semantic affinity into both graph construction and propagation, yielding a more stable and scene-consistent topology for remote sensing imagery and enabling robust feature aggregation under complex textures and cross-scale structures. Fractal-domain modeling captures scale-consistent structural statistics (e.g., self-similarity and singularity), rather than merely aggregating features across resolutions [21], [28]. This enables graph construction to better align with intrinsic data geometry, leading to more robust representations in cluttered and non-stationary remote sensing scenes.

III. FD-ViG ARCHITECTURE AND THEORETICAL ANALYSIS

As illustrated in Fig. 2, the overall pipeline starts from an input image and proceeds through three stages: *Fractal-Domain*

Learning, Fractal Graph Construction, and Graph Propagation. The model contains three core modules:

- 1) *Fractal-Domain Learning Module*: extracts multi-scale fractal features and learnable embeddings. It computes local Hölder exponents and SPSPD to describe patch-level texture complexity and energy distribution, while a parallel semantic branch encodes appearance and positional embeddings and then fuses them with fractal features to obtain unified node representations.
- 2) *Fractal Graph Construction Module*: constructs topology using multiple cues. It measures fractal similarity to capture texture-level relations, and combines embedding similarity and geometric proximity constraints to generate a weighted adjacency matrix that better aligns with scene semantic organization.
- 3) *Graph Propagation Module*: performs power-law weighted multi-scale message passing and aggregation on the constructed graph to integrate cross-region dependencies and global structure adaptively.

A. Fractal-Domain Learning Module

1) *Fractal Adaptor*: As shown in Fig. 2, the *Fractal Adaptor* implements the fractal-domain feature extraction pipeline, which converts each image patch into a compact SPSPD-based fractal descriptor via local Hölder exponent analysis. Given an input image

$$\mathbf{I} \in \mathbb{R}^{B \times C \times H \times W},$$

where B is the batch size, $C = 3$ is the number of channels, and H, W are the spatial dimensions, we partition it into non-overlapping patches of size $P \times P$ with

$$N = (H/P) \times (W/P).$$

Each patch is first embedded by a learnable projection $\Phi_{\text{emb}}(\cdot)$:

$$\mathbf{Z}_{\text{patch}} = \Phi_{\text{emb}}(\mathbf{I}) \in \mathbb{R}^{B \times N \times D}. \quad (1)$$

The Fractal Adaptor consists of the following two steps.

The full procedure of the Fractal Adaptor is summarized in Algorithm 1. It first estimates the local Hölder exponent map and then accumulates SPSPD energy by α -binning to form a compact fractal descriptor for each patch.

a) *Local Hölder exponent estimation*: For each pixel (u, v) , we define a set of multi-scale kernels $\{w_\ell\}_{\ell=1}^L$ and compute scale-dependent local responses

$$U_\ell(u, v) = \text{Pool}_{w_\ell}(\mathbf{x})(u, v), \quad (2)$$

where $\mathbf{x} \in \mathbb{R}^{C \times H \times W}$ and $\text{Pool}_{w_\ell}(\cdot)$ denotes a local aggregation operator (implemented as local max pooling). A power-law relation in the log domain is fitted as

$$\log U_\ell(u, v) = h(u, v) \log w_\ell + C(u, v), \quad (3)$$

where $h(u, v)$ is the Hölder exponent. It is estimated by least squares:

$$h(u, v) = \frac{\sum_\ell (\log w_\ell - \bar{w})(\log U_\ell(u, v) - \bar{U})}{\sum_\ell (\log w_\ell - \bar{w})^2}, \quad (4)$$

Algorithm 1: Fractal Adaptor: SPSPD-based Fractal Descriptor Extraction.

Require: $\mathbf{I} \in \mathbb{R}^{B \times C \times H \times W}$, patch grid P_n , scales $\{w_\ell\}_{\ell=1}^L$, bins $\{\alpha_b\}_{b=0}^{N_\alpha-1}$, Δ , projection W_f
Ensure: $Z_{\text{frac}} \in \mathbb{R}^{B \times N \times D}$, $\text{aux} \in \mathbb{R}^{B \times N}$
1: Patchify $\mathbf{I}$ into $\{\mathbf{x}_i\}_{i=1}^{BN}$, $\mathbf{x}_i \in \mathbb{R}^{C \times p_{sh} \times p_{sw}}$, $N = P_n^2$
2: **for** each patch $\mathbf{x}_i$ **do**
3: Compute Hölder map H_i by LS fitting:
 $\log U_\ell(u, v) = h_i(u, v) \log w_\ell + C_i(u, v)$ with
 $U_\ell = \text{Pool}_{w_\ell}(\mathbf{x}_i)$
4: Build SPSPD: $S_{i,b} \leftarrow \sum_{c,u,v} (x_{i,c}(u, v))^2 \cdot \mathbf{1}\{\alpha_b \leq h_i(u, v) < \alpha_b + \Delta\}$, $b = 0, \dots, N_\alpha - 1$
5: Form spectrum tensor $\hat{H}_i \in \mathbb{R}^{C \times N_\alpha}$, then apply
 $\hat{H}_i \leftarrow \log(1 + \text{softplus}(\hat{H}_i))$
6: Normalize $\hat{H}_i$ over (C, N_α) to get $\tilde{H}_i$; flatten
 $\mathbf{g}_i = \text{vec}(\tilde{H}_i) \in \mathbb{R}^{CN_\alpha}$
7: Tokenize $z_i \leftarrow \text{LN}(W_f \mathbf{g}_i) \in \mathbb{R}^D$; $\text{aux}_i \leftarrow \max(\mathbf{g}_i)$
8: **end for**
9: Reshape $\{z_i\}$ and $\{\text{aux}_i\}$ to $Z_{\text{frac}} \in \mathbb{R}^{B \times N \times D}$ and
 $\text{aux} \in \mathbb{R}^{B \times N}$
10: **return** Z_{frac} , aux

yielding a Hölder exponent map $H = \{h(u, v)\}$ that characterizes local roughness and irregularity.

b) *Singularity power spectrum density (SPSPD)*: Based on H , we summarize how signal energy distributes over singularity exponents. Let $[\alpha_{\min}, \alpha_{\max}]$ be uniformly divided into N_α bins of width Δ , and define

$$\mathbb{I}_b(u, v) = \begin{cases} 1, & \alpha_b \leq h(u, v) < \alpha_b + \Delta, \\ 0, & \text{otherwise,} \end{cases} \quad (5)$$

where $b = 0, 1, \dots, N_\alpha - 1$. The singularity energy in bin b is accumulated as

$$S_b = \sum_{c=1}^C \sum_{u,v} (x_c(u, v))^2 \mathbb{I}_b(u, v), \quad (6)$$

and each patch is represented by an SPSPD vector

$$h_i = [S_0, S_1, \dots, S_{N_\alpha-1}]^\top \in \mathbb{R}^{N_\alpha}. \quad (7)$$

The resulting SPSPD vectors $\{h_i\}$ are then projected to the embedding space to form the fractal feature matrix $Z_{\text{frac}} \in \mathbb{R}^{N \times D}$, which is fused with semantic patch embeddings in the subsequent Fusion module.

Algorithm 1 highlights that the core computations lie in the Hölder estimation step (Line 3) and the SPSPD energy accumulation with α -binning (Line 4). The spectrum stabilization and normalization (Lines 5–6) improve numerical robustness before the final tokenization (Line 7), producing Z_{frac} for the subsequent fusion module.

2) *Fractal Feature Mapping and Fusion*: Each SPSPD vector h_i produced by the Fractal Adaptor is projected into the D -dimensional embedding space:

$$z_i = \text{LN}(W_f h_i), \quad (8)$$

where $W_f \in \mathbb{R}^{D \times N_\alpha}$ and $\text{LN}(\cdot)$ denotes LayerNorm. Collecting all patches yields the fractal feature matrix

$$Z_{\text{frac}} = [z_1, z_2, \dots, z_N] \in \mathbb{R}^{N \times D}.$$

To integrate semantic patch embeddings $X_{\text{patch}} \in \mathbb{R}^{N \times D}$ with fractal-domain representations, we employ a gating-based fusion mechanism:

$$\text{gate} = \sigma(W_g[X_{\text{patch}} \parallel Z_{\text{frac}}]), \quad (9)$$

$$X_0 = \text{gate} \odot X_{\text{patch}} + (1 - \text{gate}) \odot Z_{\text{frac}}, \quad (10)$$

where $\sigma(\cdot)$ is the sigmoid function, $[\cdot \parallel \cdot]$ denotes concatenation, and $\odot$ is element-wise multiplication. The fused representation X_0 serves as the node feature matrix for subsequent fractal graph construction.

B. Fractal Graph Construction Module

We adaptively construct a graph in the fused feature space by jointly modeling semantic dependence and fractal similarity.

We first project X_0 into query, key, and value spaces:

$$Q = X_0 W_Q, \quad K = X_0 W_K, \quad V = X_0 W_V, \quad (11)$$

where $W_Q, W_K, W_V \in \mathbb{R}^{D \times D}$. Semantic attention similarity is

$$S_{\text{sem}} = \text{softmax}\left(\frac{QK^\top}{\sqrt{D}}\right) \in \mathbb{R}^{N \times N}. \quad (12)$$

In the fractal feature space, we compute a fractal similarity matrix:

$$S_{\text{frac}} = \text{softmax}\left(\frac{Z_{\text{frac}} Z_{\text{frac}}^\top}{\tau_f}\right), \quad (13)$$

where $\tau_f > 0$ is a temperature. In our implementation, τ_f is set to 0.85. Smaller τ_f yields sharper (more concentrated) adjacency, while larger τ_f produces smoother weights.

We combine both similarities:

$$S = (1 - \lambda)S_{\text{sem}} + \lambda S_{\text{frac}}, \quad 0 \leq \lambda \leq 1. \quad (14)$$

Then we sparsify with Top- K per node to obtain adjacency $A \in \mathbb{R}^{N \times N}$:

$$A_{ij} = \begin{cases} S_{ij}, & \text{if } S_{ij} \in \text{Top-}K(S_i), \\ 0, & \text{otherwise,} \end{cases} \quad i, j = 1, \dots, N. \quad (15)$$

We then symmetrize the graph by $A \leftarrow \frac{1}{2}(A + A^\top)$, add self-loops, and apply the standard symmetric normalization:

$$\tilde{A} = D^{-\frac{1}{2}}(A + I)D^{-\frac{1}{2}}, \quad (16)$$

where D is the degree matrix of $(A + I)$, i.e., $D_{ii} = \sum_j (A_{ij} + I_{ij})$. This normalization makes $\tilde{A}$ symmetric and ensures $\|\tilde{A}\|_2 \leq 1$, which facilitates stability analysis of power-law diffusion. The resulting $\tilde{A}$ encodes both semantic affinity and fractal self-similarity.

C. Graph Propagation Module

Standard multi-layer neighborhood aggregation often suffers from over-smoothing and over-diffusion. Moreover, fixed-hop

Algorithm 2: Power-law Graph Propagation Block.

Require: Node features $X^{(t)} \in \mathbb{R}^{N \times D}$, normalized adjacency $\tilde{A}$, order K , exponent p , projections Θ, Θ_2

Ensure: Updated features $X^{(t+1)}$

1: $X^{(0)} \leftarrow X^{(t)}$

2: **for** $k = 1$ to K **do**

3: $X^{(k)} \leftarrow \tilde{A}X^{(k-1)}$

4: **end for**

5: Compute weights $\alpha_0 \leftarrow 1$, $\alpha_k \leftarrow (k+1)^{-p}$ for $k = 1, \dots, K$, and normalize $\bar{\alpha}_k = \alpha_k / \sum_{j=0}^K \alpha_j$

6: $\tilde{X} \leftarrow \sum_{k=0}^K \bar{\alpha}_k X^{(k)}$

7: $H^{(t)} \leftarrow \Theta_2(\text{GELU}(\Theta(\tilde{X})))$

8: $X^{(t+1)} \leftarrow \text{LN}(X^{(t)} + \text{DropPath}(H^{(t)}))$

9: $X^{(t+1)} \leftarrow X^{(t+1)} + \text{MLP}(\text{LN}(X^{(t+1)}))$

10: **return** $X^{(t+1)}$

propagation can introduce scale bias. To address this, we adopt a power-law weighted multi-scale diffusion mechanism.

Given the normalized adjacency matrix $\tilde{A}$, we perform a power-law weighted multi-scale diffusion with shared projections. Starting from the current node features $X^{(t)}$, we generate K -hop propagated features recursively:

$$X^{(0)} = X^{(t)}, \quad X^{(k)} = \tilde{A}X^{(k-1)}, \quad k = 1, 2, \dots, K. \quad (17)$$

We then aggregate multi-hop information using a normalized power-law kernel with unified coefficients. Define

$$\alpha_0 = 1, \quad \alpha_k = (k+1)^{-p}, \quad k \geq 1, \quad p > 0, \quad (18)$$

and the normalized aggregation

$$\bar{X} = \frac{\sum_{k=0}^K \alpha_k X^{(k)}}{\sum_{j=0}^K \alpha_j}. \quad (19)$$

We apply shared projections and nonlinearity:

$$H^{(t)} = \Theta_2(\text{GELU}(\Theta(\bar{X}))), \quad (20)$$

where $\Theta(\cdot)$ and $\Theta_2(\cdot)$ are learnable linear projections shared across all propagation orders. The exponent p controls the scale preference: larger p emphasizes local aggregation, while smaller p increases long-range diffusion. The forward computation of the proposed power-law propagation block is summarized in Algorithm 2.

The key steps of Algorithm 2 are the recursive K -hop diffusion (Lines 2–4) and the normalized power-law aggregation (Lines 5–6), which balance local and long-range messages via the exponent p . The shared projections with nonlinearity (Line 7) yield the graph-aware update, which is finally wrapped by residual normalization and an MLP (Lines 8–9).

Each graph propagation block follows a Transformer-style design, where the above graph-aware propagation is followed by a position-wise FFN (two-layer MLP) with residual connections, formulated as:

$$X^{(t+1)} = \text{LN}\left(X^{(t)} + \text{DropPath}\left(H^{(t)}\right)\right), \quad (21)$$

$$X^{(t+1)} = X^{(t+1)} + \text{MLP}\left(\text{LN}(X^{(t+1)})\right). \quad (22)$$

TABLE I
COMPARISON WITH GRAPH-BASED MODELS ON THREE REMOTE SENSING DATASETS

Method	Params (M)	UCM	RSSCN7	SIRI-WHU
GCN [11]	0.30	62.62	54.35	56.88
GAT [12]	0.35	68.33	57.17	60.63
GraphSAGE [2]	0.32	69.29	58.04	62.70
HGNN [15]	0.38	71.19	58.70	64.38
ViG [5]	3.20	79.29	71.52	70.63
WiGNet [38]	2.85	90.16	88.03	91.04
ViHGNN [39]	4.47	86.89	88.04	89.85
Ours (FD-ViG)	2.60	91.75 ($\uparrow 1.59$)	89.52 ($\uparrow 1.48$)	92.78 ($\uparrow 1.74$)

Note: $\uparrow$ indicates the accuracy improvement over the best baseline in this group.

After T layers, we apply global average pooling to obtain an image-level representation:

$$f = \frac{1}{N} \sum_{i=1}^N X_i^{(T)}, \quad \mathbf{o} = W_c f, \quad (23)$$

where W_c is the classifier weight matrix and $\mathbf{o}$ denotes the output logits. We optimize the cross-entropy loss over logits $\mathbf{o}$.

D. Theoretical Analysis of FD-ViG

We provide a theoretical interpretation of fractal-based affinity from the perspective of structured priors and multi-scale geometry.

1) *Weak Fractal Assumption*: The proposed formulation does not require strict mathematical fractals. Instead, it relies on an approximate self-similarity property across limited scale ranges, referred to as the *weak fractal* assumption. Such behavior has been widely observed in natural images and remote sensing data, where local structures exhibit scale-consistent patterns [29]. This property enables fractal descriptors to capture stable structural characteristics that are less sensitive to noise and local perturbations.

2) *Fractal-induced Affinity Prior*: Let the image be partitioned into N patches, and denote the fused node feature matrix as $X \in \mathbb{R}^{N \times D}$, where the i -th row $\mathbf{h}_i \in \mathbb{R}^D$ represents the feature of the i -th patch (i.e., $X = X_0$ in our model). For each patch, we compute a fractal descriptor, denoted as the corresponding row in Z_{frac} , which summarizes multi-scale texture statistics.

We define a fractal-induced affinity prior as

$$A_{ij}^{(f)} = \frac{\exp(\kappa(Z_{\text{frac},i}, Z_{\text{frac},j})/\tau_f)}{\sum_k \exp(\kappa(Z_{\text{frac},i}, Z_{\text{frac},k})/\tau_f)}, \quad (24)$$

where $\kappa(\cdot, \cdot)$ denotes a similarity function. In practice, $\kappa(\cdot, \cdot)$ is instantiated as dot-product similarity, and τ_f is a temperature parameter.

The resulting matrix $A^{(f)} \in \mathbb{R}^{N \times N}$ defines a fully-connected affinity graph induced by fractal similarity. This further leads to a fractal graph Laplacian $\mathbf{L}_f = \mathbf{D}_f - \mathbf{A}^{(f)}$, which induces a smoothness prior:

$$\mathcal{R}_f(X) = \frac{1}{2} \sum_{i,j} A_{ij}^{(f)} \|\mathbf{h}_i - \mathbf{h}_j\|_2^2 = \text{tr}(X^\top \mathbf{L}_f X). \quad (25)$$

This regularization encourages nodes with similar fractal statistics to have consistent representations.

3) *Optimization and Propagation Interpretation*: The learning objective can be written as

$$\min_X \mathcal{L}_{cls}(X) + \lambda \mathcal{R}_f(X), \quad (26)$$

where λ controls the strength of the fractal prior. A gradient descent step on $\mathcal{R}_f$ yields

$$X^{(t+1)} \approx (1 - \alpha)X^{(t)} + \alpha \mathbf{A}^{(f)} X^{(t)}, \quad (27)$$

which shows that graph aggregation with $\mathbf{A}^{(f)}$ can be interpreted as optimizing the fractal smoothness prior. This provides a theoretical connection between fractal affinity and graph propagation, and can be viewed as a first-order approximation of the proposed multi-scale power-law diffusion.

4) *Information-Theoretic Perspective*: Fractal descriptors aggregate multi-scale variations into compact statistical summaries, which can be interpreted as reducing representation uncertainty in an approximate sense. By encoding scale-consistent structures, they suppress high-frequency noise and improve the stability of similarity estimation, leading to more reliable affinity modeling.

5) *Bayesian and Manifold Perspective*: From a Bayesian viewpoint, fractal representations act as a structured prior over the feature space, constraining features to lie on a scale-consistent low-dimensional manifold. This reduces the effective hypothesis space and improves robustness under noise, variability, and limited data.

6) *Practical Graph Construction*: It is important to note that $A^{(f)}$ represents a *fractal-induced affinity prior* rather than the final adjacency matrix. In practice, this prior corresponds to the fractal similarity matrix S_{frac} , which is further combined with semantic similarity and sparsified to construct the final graph. The normalized adjacency $\tilde{A}$ used in propagation can therefore be viewed as an extension of this prior under additional semantic and structural constraints.

Based on the above formulation, the proposed graph construction can be interpreted as a practical instantiation of the fractal-induced affinity prior, where fractal similarity, attention, and sparsification jointly produce a data-adaptive graph structure aligned with intrinsic data geometry.

IV. CONVERGENCE AND ERROR ANALYSIS

A. Consistency of SPSPD and Finite-Sample Error Bounds

This section studies statistical consistency of the Singularity Power Spectrum Density (SPSPD), which provides stable fractal-domain inputs for graph propagation. Since SPSPD is derived from Hölder exponent estimation based on finite scales (e.g., $3 \times 3, 5 \times 5, 7 \times 7$) and power-law fitting, each SPSPD can be viewed as a finite-sample estimate from an underlying distribution.

1) *Modeling assumption (class-conditional distribution)*: For a given class z , we view remote sensing images as sample paths generated by a class-conditional fractal random field $H_z(x)$, which exhibits stable scale-law statistics within a scale range. Under common assumptions such as local stationarity and ergodicity, spatial sampling is approximately i.i.d. in distributional terms [30]. Thus, images of class z are different realizations, and the set of local Hölder exponents extracted from an image can be approximated as finite samples from a class-conditional distribution F_z . Inter-image variations are attributed to finite-sample fluctuations and observation noise, consistent with classic texture modeling as locally stationary random fields [31] and the widely used “same-class patch distribution” assumption [32], [33].

Let $\mathcal{H}_z$ be the class-conditional random variable of Hölder exponents with i.i.d. samples $\{\mathcal{H}_{z,i}\}_{i=1}^N$. Define the population cumulative distribution function (CDF) and the empirical CDF:

$$F_z(\alpha) = \mathbb{P}(\mathcal{H}_z \leq \alpha), \quad \hat{F}_z(\alpha) = \frac{1}{N} \sum_{i=1}^N \mathbf{1}_{\{\mathcal{H}_{z,i} \leq \alpha\}}. \quad (28)$$

We further view SPSPD as a continuous functional ψ of the distribution:

$$P_z(\alpha) = \psi(F_z)(\alpha), \quad \hat{P}_z(\alpha) = \psi(\hat{F}_z)(\alpha). \quad (29)$$

Theorem IV.1 (Uniform almost sure consistency of SPSPD): Assume that for class z , the population CDF is F_z and the empirical CDF is $\hat{F}_z$. Let $P_z(\alpha)$ and $\hat{P}_z(\alpha)$ be the population and empirical SPSPD defined in Eq. (X), where ψ is continuous on $F_z(\mathcal{I})$ and $\mathcal{I} \subset \mathbb{R}$ is a compact interval. Then

$$\sup_{\alpha \in \mathcal{I}} |\hat{P}_z(\alpha) - P_z(\alpha)| \xrightarrow{a.s.} 0, \quad (30)$$

i.e., the empirical SPSPD converges uniformly almost surely to the population SPSPD over $\mathcal{I}$.

Proof: By the Glivenko–Cantelli theorem [34], the empirical CDF converges uniformly almost surely:

$$\sup_{\alpha \in \mathbb{R}} |\hat{F}_z(\alpha) - F_z(\alpha)| \xrightarrow{a.s.} 0. \quad (31)$$

In particular, the same holds over any compact interval $\mathcal{I}$:

$$\sup_{\alpha \in \mathcal{I}} |\hat{F}_z(\alpha) - F_z(\alpha)| \xrightarrow{a.s.} 0. \quad (32)$$

Since $F_z(\mathcal{I})$ is compact and ψ is continuous on $F_z(\mathcal{I})$, by the Heine–Cantor theorem [35], ψ is uniformly continuous on this set. Thus, for any $\varepsilon > 0$, there exists $\delta > 0$ such that for all $x, y \in F_z(\mathcal{I})$, $|x - y| < \delta$ implies $|\psi(x) - \psi(y)| < \varepsilon$.

By uniform convergence of $\hat{F}_z$ to F_z , there exists a (random) $N_0(\omega)$ such that for all $N \geq N_0(\omega)$,

$$\sup_{\alpha \in \mathcal{I}} |\hat{F}_z(\alpha) - F_z(\alpha)| < \delta \quad \text{a.s.} \quad (33)$$

Hence for all $\alpha \in \mathcal{I}$, we have $|\hat{F}_z(\alpha) - F_z(\alpha)| < \delta$, and uniform continuity yields

$$|\psi(\hat{F}_z(\alpha)) - \psi(F_z(\alpha))| < \varepsilon. \quad (34)$$

Taking the supremum over $\alpha \in \mathcal{I}$ gives

$$\sup_{\alpha \in \mathcal{I}} |\hat{P}_z(\alpha) - P_z(\alpha)| = \sup_{\alpha \in \mathcal{I}} |\psi(\hat{F}_z(\alpha)) - \psi(F_z(\alpha))| \leq \varepsilon. \quad (35)$$

Since ε is arbitrary, the claim follows.

Remark IV.2 (Interpretation of F_z): F_z is the *class-conditional* distribution of Hölder exponents for class z , capturing statistical fractal regularities at the class level rather than a deterministic “true spectrum” for a single image. Each image provides only finite samples (local Hölder exponents) from F_z . Therefore, Theorem IV.1 establishes uniform consistency of the SPSPD estimator as the sample size increases, without requiring identical spectra for every image instance.

2) *Finite-sample uniform error bound*: Assume ψ is Lipschitz on $F_z(\mathcal{I})$ with constant $L > 0$, i.e., $|\psi(x) - \psi(y)| \leq L|x - y|$.

Corollary IV.3 (Finite-sample uniform error bound): Under the setting of Theorem IV.1, if $\{\mathcal{H}_{z,i}\}_{i=1}^N$ are i.i.d. samples, then for any $\delta \in (0, 1)$,

$$\mathbb{P}\left(\sup_{\alpha \in \mathcal{I}} |\hat{P}_z(\alpha) - P_z(\alpha)| \leq L\sqrt{\frac{1}{2N} \ln \frac{2}{\delta}}\right) \geq 1 - \delta. \quad (36)$$

Proof: For 1D i.i.d. samples, the Dvoretzky–Kiefer–Wolfowitz inequality [36] gives

$$\mathbb{P}\left(\sup_{\alpha \in \mathbb{R}} |\hat{F}_z(\alpha) - F_z(\alpha)| \leq \varepsilon\right) \geq 1 - 2e^{-2N\varepsilon^2}. \quad (37)$$

Set $\varepsilon = \sqrt{\frac{1}{2N} \ln \frac{2}{\delta}}$. Then $\sup_{\alpha \in \mathcal{I}} |\hat{F}_z(\alpha) - F_z(\alpha)| \leq \varepsilon$ holds with probability at least $1 - \delta$. On this event, the Lipschitz property yields

$$\sup_{\alpha \in \mathcal{I}} |\hat{P}_z(\alpha) - P_z(\alpha)| \leq L\varepsilon = L\sqrt{\frac{1}{2N} \ln \frac{2}{\delta}}. \quad (38)$$

The bound indicates that the SPSPD estimation error decays at the same $O(N^{-1/2})$ rate as the empirical CDF in the sup norm [36].

B. Stability and Tail Control of Power-Law Propagation

This section analyzes numerical stability of the power-law propagation in FD-ViG under the *shared-projection* implementation, including boundedness of the propagation operator, optional contraction convergence under an additional condition, and tail control for truncated multi-hop aggregation.

1) *Setup and notation*: Let $\tilde{A} \in \mathbb{R}^{n \times n}$ be the normalized propagation matrix with spectral norm

$$\|\tilde{A}\|_2 \leq r \leq 1. \quad (39)$$

Let Θ and Θ_2 be the shared linear transforms used after multi-hop aggregation, and assume

$$\|\Theta\|_2 \leq \beta, \quad \|\Theta_2\|_2 \leq \beta_2. \quad (40)$$

Define unified power-law coefficients

$$\alpha_0 = 1, \quad \alpha_k = (k+1)^{-p}, \quad k \geq 1, \quad (41)$$

and the normalized truncated aggregation operator

$$M_K = \frac{\sum_{k=0}^K \alpha_k \tilde{A}^k}{\sum_{j=0}^K \alpha_j}, \quad p > 0. \quad (42)$$

Let $\sigma(\cdot)$ denote the point-wise nonlinearity (GELU). Since σ is smooth, it is Lipschitz on any bounded domain. We denote its Lipschitz constant on the relevant feature range by L_σ .

2) *Propagation operator and boundedness*: The graph propagation update in our implementation can be written as the operator

$$\mathcal{T}(X) = \Theta_2(\sigma(\Theta(M_K X))). \quad (43)$$

We first bound $\|M_K\|_2$. Using $\|\tilde{A}^k\|_2 \leq r^k$ and the triangle inequality,

$$\|M_K\|_2 \leq \frac{\sum_{k=0}^K \alpha_k \|\tilde{A}^k\|_2}{\sum_{j=0}^K \alpha_j} \leq \frac{\sum_{k=0}^K \alpha_k r^k}{\sum_{j=0}^K \alpha_j} \leq 1. \quad (44)$$

Therefore, for any X in the bounded feature range,

$$\begin{aligned} \|\mathcal{T}(X)\|_2 &\leq \|\Theta_2\|_2 L_\sigma \|\Theta\|_2 \|M_K\|_2 \|X\|_2 \\ &\leq \gamma \|X\|_2, \end{aligned} \quad (45)$$

where $\gamma := \beta_2 L_\sigma \beta$. Thus $\mathcal{T}$ is a bounded (Lipschitz) operator on the relevant feature domain, which is sufficient to prevent numerical blow-up in finite-depth networks.

3) *Optional contraction convergence*: For the unbiased iteration $X^{(l+1)} = \mathcal{T}(X^{(l)})$, any finite depth L satisfies

$$\|X^{(L)}\|_2 \leq \gamma^L \|X^{(0)}\|_2. \quad (46)$$

If additional regularization or residual scaling enforces $\gamma < 1$, then $\mathcal{T}$ becomes a contraction, and Banach's fixed-point theorem [35] implies

$$\|X^{(l)}\|_2 \leq \gamma^l \|X^{(0)}\|_2 \xrightarrow{l \rightarrow \infty} 0. \quad (47)$$

This provides an optional strong convergence guarantee, although it is not required for finite-depth stability.

4) *Biased propagation*: If a bias term exists,

$$X^{(l+1)} = \mathcal{T}(X^{(l)}) + B, \quad (48)$$

finite-depth outputs remain bounded when $\gamma < \infty$; if $\gamma < 1$, there exists a unique fixed point

$$X^* = (I - \mathcal{T})^{-1} B, \quad (49)$$

and

$$\|X^{(l)} - X^*\|_2 \leq \gamma^l \|X^{(0)} - X^*\|_2 \rightarrow 0. \quad (50)$$

5) *Tail control (truncation error)*: Consider the infinite-order normalized operator

$$M_\infty = \frac{\sum_{k=0}^{\infty} \alpha_k \tilde{A}^k}{\sum_{j=0}^{\infty} \alpha_j}, \quad \alpha_0 = 1, \quad \alpha_k = (k+1)^{-p} \quad (k \geq 1), \quad (51)$$

and its truncated counterpart

$$M_K = \frac{\sum_{k=0}^K \alpha_k \tilde{A}^k}{\sum_{j=0}^K \alpha_j}, \quad K \in \mathbb{N}. \quad (52)$$

For $p > 1$, $\sum_{k \geq 0} \alpha_k < \infty$, hence M_∞ is well-defined.

Define the truncation error operator $\Delta_K := M_\infty - M_K$. Using $\|\tilde{A}^k\|_2 \leq 1$ and $\sum_{j=0}^K \alpha_j \leq \sum_{j=0}^{\infty} \alpha_j$, we obtain

$$\left\| \sum_{k > K} \alpha_k \tilde{A}^k \right\|_2 \leq \sum_{k > K} \alpha_k. \quad (53)$$

A standard normalization-difference argument yields

$$\|\Delta_K X\|_2 \leq C \left(\sum_{k > K} \alpha_k \right) \|X\|_2, \quad (54)$$

where $C > 0$ is independent of K .

Finally, by the integral test [37], for $p > 1$,

$$\sum_{k > K} (k+1)^{-p} = O(K^{1-p}), \quad (55)$$

and therefore the truncation error decays polynomially:

$$\|\Delta_K X\|_2 = \|(M_\infty - M_K)X\|_2 = O(K^{1-p}) \|X\|_2. \quad (56)$$

Overall, this result shows that the proposed power-law diffusion admits a controllable approximation: for any fixed exponent $p > 1$, the infinite-order operator M_∞ can be well-approximated by a finite-order truncation M_K with an error decaying as $O(K^{1-p})$. This guarantees that FD-ViG achieves both numerical stability at finite depth and theoretically justified long-range diffusion with bounded truncation error.

V. EXPERIMENTS AND ANALYSIS

A. Experimental Settings

We evaluate FD-ViG on three representative optical remote sensing scene classification datasets:

- 1) *UCMerced Land Use*: 21 classes, 100 RGB images per class (2100 total), focusing on fine-grained discrimination of textures and structures.
- 2) *RSSCN7*: 7 classes, 400 images per class (2800 total), featuring significant scale and orientation variations for robustness evaluation.
- 3) *SIRI-WHU*: 12 urban scene classes, about 2400 images in total, emphasizing generalization under complex urban layouts and textures.

All images are resized to 256×256 and split into 70%/15%/15% for training/validation/testing with stratified sampling. Evaluation is conducted using overall accuracy (Acc) and the number of parameters (Params).

Experiments are conducted on an NVIDIA RTX 2080Ti GPU. We use AdamW with initial learning rate 4×10^{-4} and weight

TABLE II
COMPARISON WITH CONVOLUTIONAL AND TRANSFORMER-BASED BASELINES

Method	Params (M)	UCM	RSSCN7	SIRI-WHU
GHA-ViT [40]	1.55	83.57	77.48	77.50
CABNN [18]	~2.8	86.74	82.83	82.70
TS-CNN [19]	~5.1	88.33	86.30	86.04
ViT [1]	3.90	89.84	90.95	91.39
ResNet-18 [41]	11.20	88.19	95.24	92.62
Ours (FD-ViG)	2.60	91.75 ($\uparrow 1.91$)	89.52	92.78 ($\uparrow 0.16$)

decay 1×10^{-2} . Cosine annealing is applied:

$$\eta_t = \eta_{\min} + \frac{1}{2}(\eta_0 - \eta_{\min})(1 + \cos(\frac{t}{T}\pi)),$$

where $\eta_{\min} = 10^{-3}\eta_0$.

Data augmentation includes random resized crop, horizontal flip, RandAugment ($N = 2, M = 9$), and random erasing (probability 0.25). Mixup with Beta(0.3, 0.3) is further used to improve diversity.

Key hyperparameters:

- *Patch/encoder*: $P = 32$, $D = 192$, number of heads = 6, encoder layers = 4;
- *Graph construction*: Top- $k = \{6, 8, 10\}$;
- *Fractal features*: number of spectrum bins $N_\alpha = 9$;
- *Power-law diffusion*: order $K = 3$, exponent p is learnable (initialized at $p = 1.5$, constrained to $[1.05, 6.0]$ to match the tail bound condition $p > 1$ in Sec. IV-B);
- *Fusion/loss*: λ is learnable and initialized as 0.05; auxiliary loss weight $w_{\text{aux}} = 0.25$ decayed by 0.96^t .

B. Performance Comparison

We compare FD-ViG with: (1) graph-based models (GCN [11], GAT [12], GraphSAGE [2], HGNN [15], ViG [5], WiGNet [38], and ViHGNN [39]); (2) Transformer and CNN baselines (ViT [1], GHA-ViT [40], ResNet-18 [41], TS-CNN [19], CABNN [18]).

1) *Results on graph-based baselines*: Table I shows that FD-ViG consistently outperforms representative graph-based baselines, including GCN, GAT, GraphSAGE, HGNN, as well as recent graph-based vision models such as ViG, WiGNet, and ViHGNN.

FD-ViG achieves the best performance across all three datasets, demonstrating consistent improvements over strong graph-based baselines.

In particular, compared with recent methods such as WiGNet and ViHGNN, FD-ViG still maintains clear performance advantages, indicating that the improvement is not limited to weaker baselines but also extends to competitive graph-based architectures.

We attribute these improvements to the fact that conventional graph pipelines either rely on fixed or weakly adaptive topology (often sensitive to texture noise and scale changes), or construct edges mainly from semantic embedding similarity. In contrast, our fractal-domain similarity provides a complementary cue that is more stable to self-similar textures and multi-scale roughness, thus producing a topology that better aligns with remote-sensing

scene organization. Combined with power-law multi-scale propagation, FD-ViG can aggregate both local fractal patterns and cross-region structural dependencies in a unified graph domain.

2) *Results on CNN/Transformer baselines and parameter efficiency*: As reported in Table II, FD-ViG achieves the best performance on UCMerced and SIRI-WHU with only 2.6 M parameters, demonstrating a favorable accuracy-complexity trade-off. Compared with ViT, FD-ViG improves UCMerced accuracy while using fewer parameters. Compared with ResNet-18, FD-ViG matches or slightly exceeds performance on SIRI-WHU with substantially fewer parameters, indicating that explicit topology modeling can compensate for the capacity of heavier convolutional backbones under complex layouts.

3) *Discussion on RSSCN7*: FD-ViG is below ResNet-18 on RSSCN7. A plausible explanation is that RSSCN7 contains strong viewpoint/scale/illumination variations where a high-capacity CNN benefits from aggressive data augmentation and strong translation equivariance. Our model is intentionally lightweight and relies on patch-level graph tokens; when the scene variation is dominated by global appearance changes rather than stable texture/structure cues, a deeper CNN may have an advantage. Nevertheless, FD-ViG remains competitive while being significantly more compact, suggesting that the proposed fractal-domain graph learning is a promising direction for parameter-efficient remote sensing recognition.

4) *Complexity and Scalability*: The proposed method introduces only limited overhead compared with the backbone. The additional cost mainly comes from top- k sparsification and sparse graph propagation, which scales linearly with the number of nodes. In practice, since nodes correspond to image patches (e.g., $N = 64$ for a 256×256 image with patch size 32), the graph size remains small, and the overall complexity is dominated by the backbone network. Moreover, with typical settings ($k \leq 16$, $K \leq 3$), the resulting sparse graph contains only around 10^3 edges, leading to negligible memory and computational overhead. For large satellite imagery, the method can be applied using tiled or sliding-window inference, ensuring good scalability.

Under a 256×256 input, the implemented FD-ViG requires approximately 203.2 M MACs (about 0.44 GFLOPs) and achieves an average inference latency of 13.18 ms per image on a single GPU, achieving a strong balance between accuracy and computational cost.

5) *Overall summary*: Overall, lightweight CNN baselines are often stable under fixed scales but can be less robust in scenes with pronounced self-similarity, scale complexity, and

TABLE III
ABLATION STUDY OF THE MAIN COMPONENTS IN FD-ViG ON THE UCMERGED DATASET (%)

Fractal Adapter	Power-law Propagation	Fractal Attention	Accuracy (%)
×	×	×	86.03
✓	×	×	89.21
×	✓	×	89.52
×	×	✓	86.35
✓	✓	×	91.11
✓	×	✓	90.48
×	✓	✓	90.01
✓	✓	✓	91.75



Fig. 4. Layer-1 adjacency visualization on UCMerced. Red boxes denote reference patches, and blue boxes denote Top- K neighbors. The learned neighborhoods remain semantically coherent across different scene categories.

fractal structural variations. FD-ViG leverages SPSPD-based fractal descriptors for texture/roughness characterization and couples them with adaptive graph construction and power-law propagation, achieving strong accuracy with a compact model size.

C. Ablation Study

To explicitly delineate the contributions of different components in FD-ViG, we conduct a comprehensive ablation study on the **UCMerced** dataset, following the same experimental setting as the main results. We isolate three key modules: the *Fractal Adapter*, the *Power-law Graph Propagation*, and the *Fractal*

Attention. All variants share the same backbone and graph construction strategy, ensuring that performance differences are solely attributed to these components.

The results are summarized in Table III. Both the Fractal Adapter and Power-law Propagation provide substantial individual gains, while Fractal Attention contributes marginally. Notably, their combination leads to a further improvement, indicating complementary effects. This demonstrates that the performance gain of FD-ViG arises from the synergy between fractal-domain representation and graph-based reasoning, rather than increased architectural complexity.

We further analyze the impact of key hyperparameters, including graph sparsity (Top- k) and fractal-related parameters. We observe that the model performance remains stable within a reasonable range of settings (e.g., Top- k in [6,10]), indicating that FD-ViG does not rely on precise hyperparameter tuning. In addition, fractal-related parameters (e.g., fusion weights) exhibit smooth convergence behavior during training, suggesting that the proposed formulation is stable and well-behaved in optimization.

D. Cross-Dataset Validation

To further evaluate the generalization ability of the proposed method under cross-domain settings, we conduct additional experiments on the **OpenSARUrban** dataset, which consists of SAR imagery and exhibits significantly different characteristics from optical datasets in terms of imaging mechanism, noise distribution, and texture patterns.

In this experiment, we construct a representative subset by selecting six urban-related classes, including *airport*, *railway*, *highway*, *highbuilding*, *skyscraper*, and *storage area*. This subset reflects typical urban scenarios while reducing extreme class imbalance in the original dataset.

Compared with optical images, SAR data typically involve stronger speckle noise, lower signal-to-clutter ratio, and more complex scattering effects, leading to substantial domain shift. Moreover, this dataset presents additional challenges such as class imbalance, relatively low spatial resolution, and less distinctive semantic boundaries, making classification more difficult and serving as a stringent testbed for evaluating representation robustness.

Following a consistent experimental setup, we compare FD-ViG with representative CNN-based, Transformer-based, and

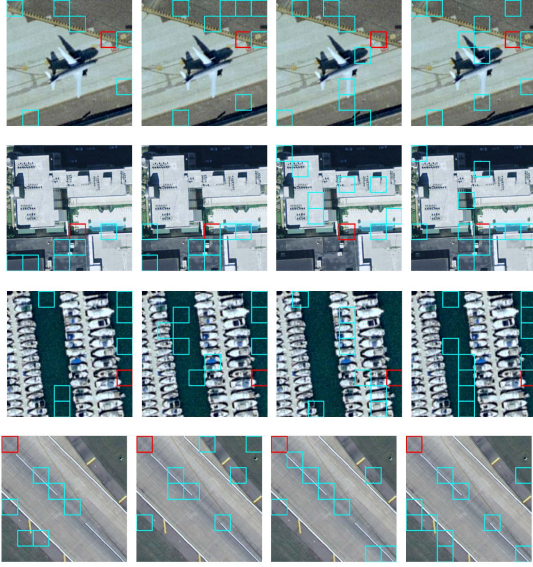


Fig. 5. Multilayer graph propagation visualization. From left to right, each column of subfigures shows the propagation process from Layer 0 to Layer 3 for the four target categories, respectively. As the depth increases, the receptive field of each node gradually expands, and deeper layers progressively aggregate into stable global topological patterns.

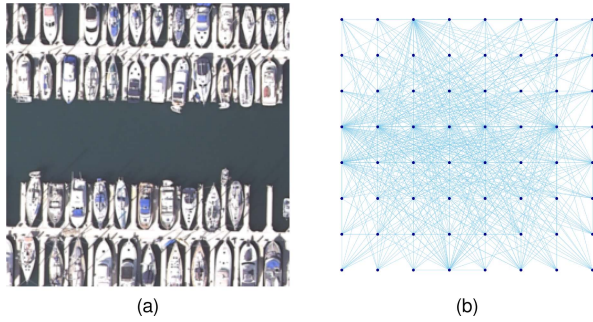


Fig. 6. Final-layer graph visualization (Harbor). The topology separates “water” and “ship” regions while preserving a small number of weak cross-region links, demonstrating adaptive structure expression after multilayer propagation.

graph-based models. The results are reported in Table IV. As shown, FD-ViG achieves the best performance among all compared methods, outperforming both classical CNN and recent graph/transformer-based approaches.

These results suggest that the proposed fractal-domain representation captures intrinsic multi-scale structural patterns that remain stable across sensing modalities. In particular, fractal descriptors emphasize scale-consistent statistics rather than raw appearance, making them less sensitive to domain-specific variations such as noise and imaging differences.

Furthermore, the graph construction based on fractal similarity aligns message passing with the underlying structural geometry, rather than relying solely on Euclidean or semantic proximity. This enables more reliable relational modeling under domain shift.

Overall, these results provide strong evidence that the performance gains of FD-ViG are not dataset-specific, but stem from its ability to model fundamental structural properties that generalize across different sensors and imaging conditions.

TABLE IV
CROSS-DATASET EVALUATION ON OPENSARURBAN

Method	Accuracy (%)
WiGNet	64.58
ViHGNN	65.98
ViG	67.92
ViT	68.95
ResNet-18	70.30
Ours (FD-ViG)	71.38 (↑1.08)

Note: ↑ indicates the accuracy improvement over the best baseline on this dataset.

E. Visualization Analysis

We provide qualitative visualizations from three aspects: (i) fractal feature ratio during training, (ii) layer-1 adjacency visualization, and (iii) multi-layer propagation evolution.

1) *Fractal feature ratio curve*: Fig. 3 plots the *fractal feature ratio* during training, defined as the average activation energy contributed by the fractal branch in the gated fusion module (i.e., the relative contribution of Z_{frac} in the fusion). This metric reflects how the model allocates representation capacity between semantic patch embeddings and SPSD-based fractal descriptors.

The curve typically exhibits three stages. (i) *Rapid increase*: in early epochs, the ratio rises quickly, indicating that the network learns to exploit fractal-domain cues as stable texture/roughness signals. (ii) *Balancing adjustment*: the ratio then fluctuates moderately, suggesting the gate is actively re-weighting fractal and semantic features to accommodate class-specific appearance and layout variations rather than collapsing to a single branch. (iii) *Stable convergence*: finally, it stabilizes around 0.75, implying that fractal features become the dominant contributor after convergence.

A stable ratio above 0.5 supports our motivation that remote sensing scene recognition benefits strongly from multi-scale self-similar texture characterization. Meanwhile, the non-saturated plateau indicates that semantic embeddings still provide complementary discriminative cues, validating the necessity of fractal-semantic fusion instead of relying on either branch alone.

2) *Layer-1 graph adjacency visualization*: We visualize the first-layer adjacency on four representative classes in UCMerced (*Airplane*, *Buildings*, *Harbor*, *Runway*). Each image shows several reference patches (red boxes) and their Top- K neighbors (blue boxes), as in Fig. 4. We observe clear semantic-consistent clustering: airplane structural regions connect strongly with each other and weakly to runway background; buildings form stable clusters around repetitive roof/ridge/shadow patterns; harbor tends to separate ships from water regions; runway exhibits chain-like connectivity along the dominant direction. These results indicate that the initial fractal graph is not merely driven by pixel similarity, but by multi-scale roughness and layout consistency.

3) *Multilayer graph propagation visualization*: To illustrate receptive field expansion, we visualize propagation from Layer 0 to Layer 3 for four representative samples (Fig. 5). Shallow layers primarily aggregate local neighborhoods, while

deeper layers gradually incorporate more distant and even global structure. As the depth increases, information can traverse multiple hops along the learned topology, enabling each node to integrate long-range context beyond its local patch vicinity. Consequently, the representation evolves from local texture cues to a more global, scene-level structural understanding, indicating an effective expansion toward a global receptive field.

4) *Final-layer graph visualization*: We further visualize the final-layer topology (Layer 3, Top-10) for a *Harbor* sample, as in Fig. 6. The graph naturally separates water and ship regions with strong intra-region connectivity, while retaining a few weak cross-region edges, reflecting an adaptive “core connections + peripheral weak links” pattern that preserves both dominant partitioning and local continuity.

VI. CONCLUSION

We proposed a fractal-domain vision graph neural network (FD-ViG) that unifies SPSPD-based fractal-domain learning with power-law multi-scale graph propagation to jointly model texture complexity and spatial structural relations in remote sensing imagery. Experiments on UC Merced, RSSCN7, and SIRI-WHU demonstrate that our model achieves strong accuracy, stability, and parameter efficiency compared with mainstream CNN, Transformer, and graph baselines.

There are still opportunities for improvement. As remote sensing resolution and data scale increase, the efficiency of fractal feature extraction and multi-layer propagation should be further optimized. Moreover, the current model focuses on static images; extending the fractal-domain graph modeling to temporal/multi-phase data and dynamic structures is a promising direction. The method is most effective for data with rich multi-scale structures, and may be less effective on overly smooth or low-texture data.

Future work includes: (1) designing more efficient fractal feature extraction and propagation for large-scale scenes; (2) extending to temporal and dynamic data to better describe land-cover changes; and (3) exploring fusion with multi-modal data (e.g., optical–SAR–hyperspectral) to improve cross-modal scene understanding.

VI. DATA AVAILABILITY

Shared source code on GitHub with <https://github.com/sjtu-xiong/Fractal-GNN>. Experimental data from open-source datasets, links provided. Aim to facilitate access, enable replication, contribute to scientific community.

ACKNOWLEDGMENT

The authors would like to express sincere gratitude to the Associate Editor and the anonymous reviewers for their valuable time, insightful comments, and constructive suggestions.

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Jiacheng Yin received the BS degree in electronic and computer engineering from the University of Michigan–Shanghai Jiao Tong University Joint Institute, Shanghai Jiao Tong University, Shanghai, China, and the MS degree in instrumentation science and technology from the School of Automation and Intelligent Sensing, Shanghai Jiao Tong University, Shanghai, China. His research interests include fractal-domain deep learning, computer vision, and radar signal processing.



Tao Zhen received the BS degree from the Beijing Institute of Technology, Beijing, China, in 2021, and the MS degree in electronic and information engineering from the Institute of Sensing and Navigation, Shanghai Jiao Tong University, Shanghai, China, in 2025. His research interests include fractal neural networks, fractal-domain deep learning, radar signal processing, and their applications in radar target detection and recognition.



Gang Xiong received the PhD degree in electronic engineering from the Nanjing University of Science and Technology (NJUT), Nanjing, China, in 2005. In 2014, he joined the School of Electronic Information and Electrical Engineering with Shanghai Jiao Tong University as an associate professor. He is currently an associate professor with the School of Automation and Intelligent Sensing and also with the Institute of Sensing and Navigation, Shanghai Jiao Tong University, Shanghai, China. His research interests include non-stationary signal processing, fractal and multifractal signal or image processing, fractal deep learning, and their applications in radar/SAR weak target detection and recognition under low signal-to-noise ratio backgrounds.



Wenxian Yu (Senior Member, IEEE) was born in Shanghai, China, in 1964. He received the BSc, MSc, and PhD degrees from the National University of Defense Technology, Changsha, China, in 1985, 1988, and 1993, respectively. In 2008, he joined the School of Electronic Information and Electrical Engineering, Shanghai Jiao Tong University, Shanghai. He is currently a distinguished professor under the Yangtze River Scholar Scheme, the vice dean of the Advanced Industrial Technology Research Institute, Shanghai, and the dean with the Academy of Information Technology and Electrical Engineering, Shanghai. In these areas, he has authored more than 260 research articles. His research interests include radar target recognition, remote sensing information processing, multisensor data fusion, and integrated navigation systems.